

CASEY SKIWAY SOUTH, Antarctica

WMO: 898090

Lat: 66.285S

Lon: 110.743E

Elev: 375

StdP: 96.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-34.0	-31.3	N/A	N/A	N/A	N/A	N/A	N/A	32.6	-12.7	30.1	-11.3	3.7	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	3.3	N/A	1.7	N/A	0.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.2	21.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.2	-2.9	-5.7	-11.1	-13.8	-17.2	-16.7	-16.6	-18.2	-16.0	-14.4	-8.9	-4.4		
	DBStd	7.18	2.56	2.90	4.72	4.70	6.22	6.18	6.26	5.54	5.73	4.85	4.30	3.68		
	HDD10.0	8101	401	440	654	714	842	801	824	874	781	757	567	447		
	HDD18.3	11142	659	673	913	964	1101	1051	1082	1132	1031	1015	817	705		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.4	5.7	6.2	6.1	6.4	7.0	8.1	11.7	7.9	8.3	7.7	6.9	6.6		
(15) Precipitation	PrecAvg	457	39	40	41	36	32	39	45	45	35	31	39	37		
	PrecMax	587	55	67	72	70	53	69	79	77	61	50	70	70		
	PrecMin	362	21	5	19	13	7	12	20	17	15	11	10	15		
	PrecStd	56	10	16	13	14	11	13	17	14	11	10	14	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	3.8	-1.7	-3.5	-3.9	-4.8	-3.7	-4.2	-1.3	-3.1	1.6	6.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	3.5	0.7	-3.3	-4.9	-6.0	-6.1	-5.8	-7.4	-5.8	-5.3	-0.5	3.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	2.0	-0.4	-4.3	-6.1	-7.0	-7.0	-7.4	-9.6	-7.7	-6.9	-2.2	1.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	0.9	-1.4	-5.2	-7.4	-8.6	-8.4	-8.8	-11.0	-8.8	-8.0	-3.4	0.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		7.7	7.3	7.8	6.8	7.2	7.1	6.8	6.9	7.0	9.3	9.0	8.6		
	5% DB	MCDBR	9.6	9.7	7.3	6.3	6.6	6.6	5.7	7.4	5.2	9.3	11.1	11.8		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.243	0.222	0.184	0.122	0.122	0.126	0.181	0.210	0.219	0.238	0.254		
	taud		2.342	2.380	2.360	2.221	1.926	1.926	1.901	2.169	2.275	2.293	2.298	2.289		
	Ebn at Noon		1012	971	896	727	388	76	335	710	893	992	1024	1024		
	Edh at Noon		107	90	70	42	14	3	13	43	73	97	111	118		
(44) All-Sky Solar Radiation	RadAvg		7.71	5.34	2.81	0.96	0.17	0.01	0.07	0.58	2.11	4.73	7.37	8.64		
	RadStd		0.34	0.29	0.15	0.07	0.01	0.00	0.00	0.04	0.10	0.14	0.26	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page